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Lightweight open source on-spacecraft machine learning with mlpack and F-Prime

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ABSTRACT

Onboard machine learning is an increasingly important topic in spaceflight systems, as machine learning and AI models are increasingly used for data processing, anomaly detection, attitude control systems, and other predictive applications. However, successfully deploying a machine learning model in a spaceflight computing environment is fraught with difficulty, as spaceflight computing environments are low-resource, but typical machine learning software solutions are implemented in languages like Python, which require an interpreter and other heavyweight dependencies. It is therefore virtually mandatory for any on-spacecraft machine learning to be written with lightweight toolkits that minimize dependencies and resource usage.

We demonstrate a solution to this problem: via the use of open source C++-based scientific computing software, including the mlpack machine learning library and Armadillo linear algebra library, we provide the first implementation of a working on-spacecraft machine learning anomaly detection system inside the F-Prime flight software framework that successfully detects non-trivial anomalous events by using a kernel density estimate on telemetry data. Our prototype is validated on physical hardware, where we physically created several anomalous situations that were all successfully detected. The code is available publicly on Github and can be used as a reference project for deploying any complex machine learning pipeline to a spacecraft via the F-Prime framework.

1 INTRODUCTION

Modern spaceflight missions produce massive amounts of data. NASA missions are now generating over 100 TB of data each day,[1] and this presents a significant challenge for data processing. Increasingly, a solution to this problem is direct on-board machine learning to process the data (as opposed to downlinking the data for ground-based processing).[2–4] But the computational resources available on low-power satellites, such as CubeSats[5] and other similar craft such as extraplanetary rovers are quite limited.

Often, the computational platforms of satellites can be somewhat esoteric. Although ARM-based platforms are likely the most common, with use in missions like SpaceCube[6] and DODONA,[7] legacy platforms are also in wide usage. For instance, the ArgoMoon mission[8] and Roman Space Telescope[9] use SPARC processor. The common BAE RAD750[10] is a PowerPC processor and is used on the Perserverance and Curiosity rovers.[9] Even platforms like MIPS are in use on missions like New Horizons.[11]

A further constraint on spaceflight computing systems is the flight software framework being used to control them. Two common choices, both supported by NASA, are F-Prime[12] and core Flight System (cFS).[13] cFS is written in C, adhering to the C99 standard. Although this allows for precise programmer control in an embedded context, it makes interoperability with non-C99 toolkits difficult. For similar reasons, F-Prime is written in C++, targeting the C++11 standard or newer. This choice comes with the same interoperability difficulties.

This variety of computing platforms, the general

lack of computing resources present on them, as well as the language limitations imposed by cFS and F-Prime present a huge barrier to practitioners wishing to perform machine learning directly on a spacecraft. Most widely-used machine learning software packages, such as `scikit-learn`,[14] TensorFlow,[15] or PyTorch[16] provide interfaces in Python. This is problematic for lightweight deployment: Python requires a language interpreter, which is inefficient and can be orders of magnitude larger than storage resources permit.[17] Even projects aimed to productionalize machine learning models present problems: efforts like XLA,[18] TensorFlow Lite, and ExecuTorch for PyTorch are cumbersome to build and link against, as they use complex build systems and have many dependencies. Vendor-specific alternatives like ARM's CMSIS-NN[19] suffer from limited vendor support, limited functionality, and cannot be deployed on other hardware.

Yet there exists a widely-used native C++ scientific computing ecosystem, made up of tools like Armadillo,[20] Eigen,[21] and xTensor[22] for linear algebra, mlpack,[23] dlib-ml,[24] and ensmallen[25] for machine learning, and numerous other libraries for related tasks. Many of these tools are intentionally lightweight, header-only, and dependency-free; thus they can be easily cross-compiled to any target architecture and deployed without burdensome dependency requirements. These libraries have been deployed widely, including for other on-board spaceflight applications.[26,27]

In this paper, we put both of these pieces together to solve the problem: we demonstrate that mlpack can be easily used inside of F-Prime to build a lightweight machine learning anomaly detector that runs directly on-board the spacecraft without the need for ground-based intervention. Following Peet et al.,[28] we deployed our solution to a Pololu Romi robot, which serves as a surrogate of a small satellite or extraplanetary rover. Our anomaly detector framework was able to successfully detect a number of complex anomalous situations that are not trivially detected with, e.g., thresholding.

Although our purposes here are to detect telemetry anomalies, our solution can be trivially adapted to other common on-board machine learning tasks, such as computer vision and image classification,[29,30] health monitoring,[31] and other tasks. Our code is publicly available for use and adaptation at `https://github.com/SterlingPeet/mlpack-FPrime-robot`, licensed under the TODO license.

2 SOFTWARE OVERVIEW

2.1 F-Prime

As mentioned earlier, F-Prime is a modular flight software framework emphasizing the reusability of individual parts of software across different missions. In F-Prime, each part of the mission is decomposed into discrete *Components* with well-defined interfaces; these serve as the fundamental building blocks for the mission software.

A single mission consists of a number of components connected in a specific way, called a *deployment*. Each F-Prime component's interface is defined as a number of input and output ports, which are defined and connected to other components. This interface and connections are specified in the Fpp domain-specific language.[?] For example, the `Svc.SystemResources` component, which monitors the state of the computing environment (e.g. RAM usage, CPU utilization), typically outputs its telemetry via an Fpp connection to the mission telemetry service, which will report the telemetry to the ground control systems.

The F-Prime framework handles all of the communication between components, and provides numerous default components and deployments. This means that for a user, implementing the mission software stack typically means taking several F-Prime-supplied off-the-shelf components and connecting them with a few custom components implemented specifically for the mission.

Once the components are implemented and the connections in the deployment are defined, an F-Prime deployment can easily be cross-compiled to any target platform, including traditional platforms like VxWorks on the RAD750, and newer platforms like ARM systems running lightweight Linux distributions (for instance, a Raspberry Pi).

Compilation and configuration is handled via CMake, which is likely the most widely-used C++ build system generator. F-Prime automatically generates the CMake configuration based on the deployment topology. A user who has implemented their flight software with F-Prime can typically build and run it with a few simple commands via the use of the helper `fprime-util` and `fprime-gds` programs:

```
# automatically configure CMake
$ fprime-util generate

# build project with CMake
$ fprime-util build

# run the software and display
```

```
# ground control interface
$ fprime-gds
```

F-Prime deployments can work even on very resource-constrained hardware; for instance, it has been successfully used with program memory size as small as 128 kB and RAM as small as 64 kB.

Further resources on F-Prime can be found in the original F-Prime paper,[12] and more details related to our deployment topology can be found in the previous work of Peet et al.,[28] whose work we have used as a starting point.

3 HARDWARE OVERVIEW

4 ARCHITECTURE

5 EXPERIMENTS

6 CONCLUSION

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